

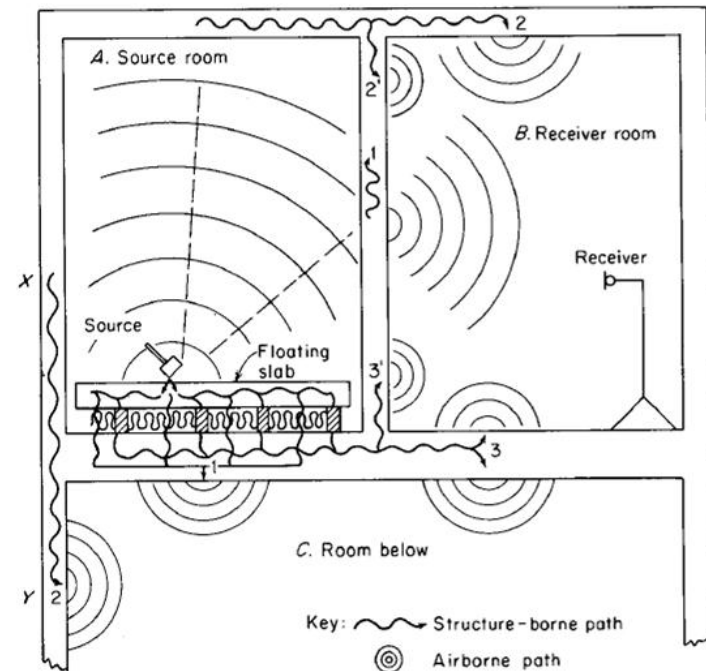
Week 13

Mechanical Noise Control

Spring 2021
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College of Architecture and Design
Lawrence Technological University

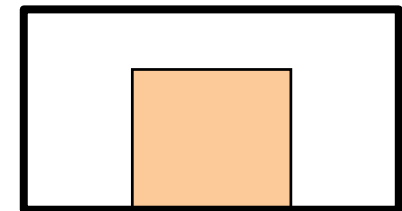
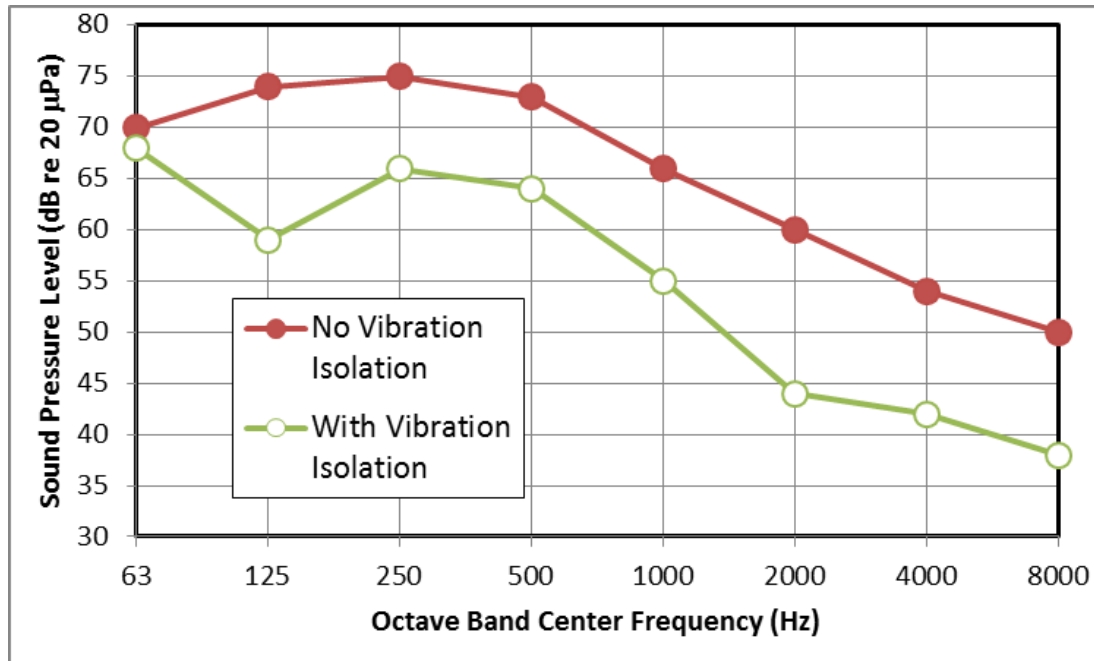
NOISE PROPAGATION

- Noise transmission can be a combination of airborne and structure-borne sound
- **Airborne sound** through walls, floors, and ceilings will be controlled by enclosure design and construction
- **Structure-borne sound** will be controlled by isolation, mass, and damping

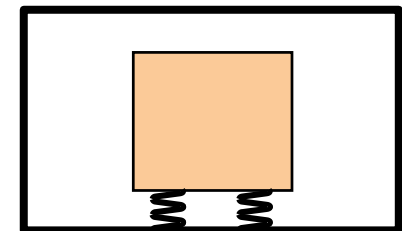


STRUCTUREBORNE SOUND PATH

- Direct contact of vibrating sources to floors, walls, ceilings and enclosures can generate significant structure-borne noise
- Isolation between source and adjoining surfaces, and eliminating contact will minimize structure-borne noise



No Isolation

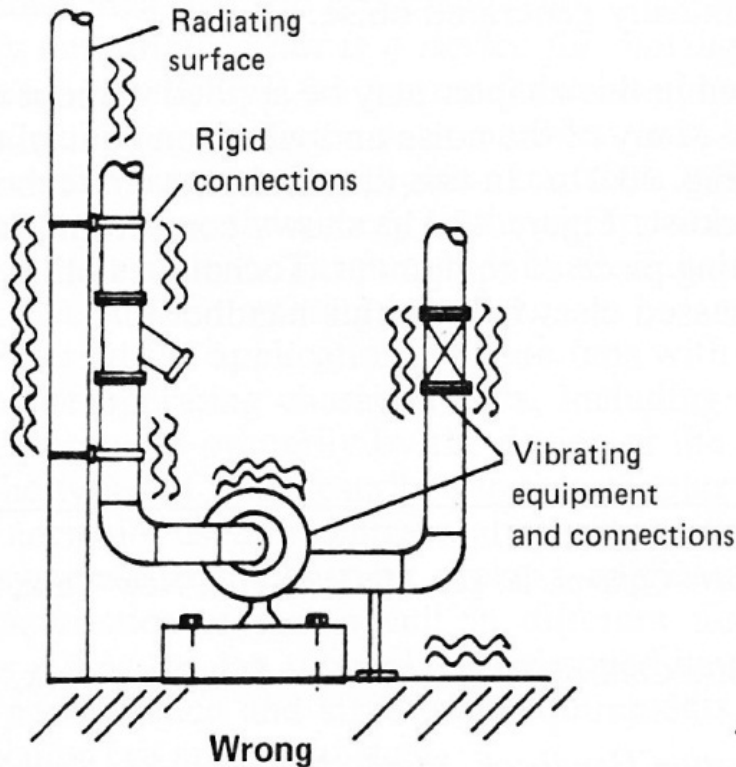


With Isolation

STRUCTUREBORNE SOUND PATH

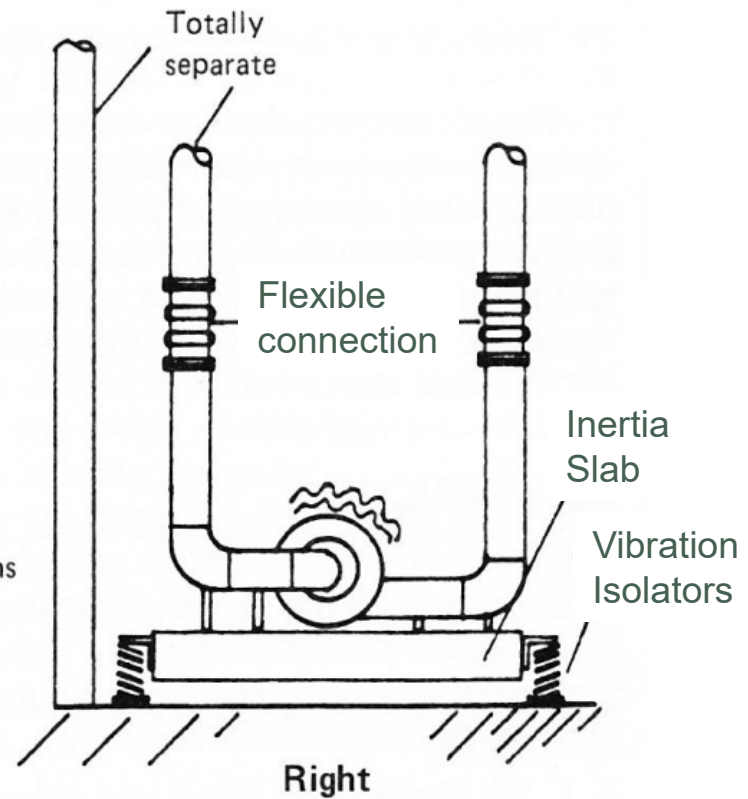
WRONG

- Hard-mounted to floor/ground
- Rigid attachment to structures
- Hard connections at pipes/ducts

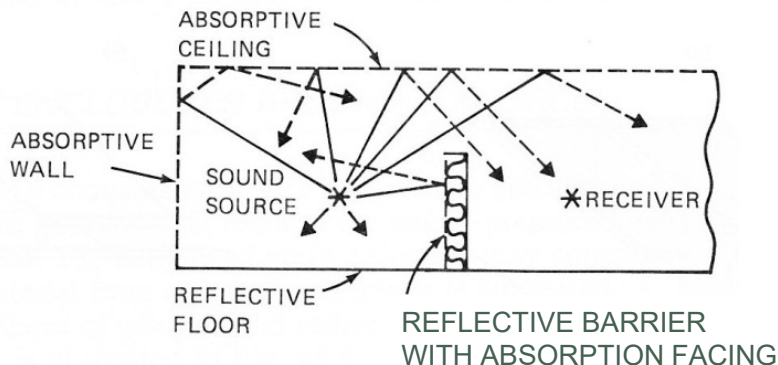
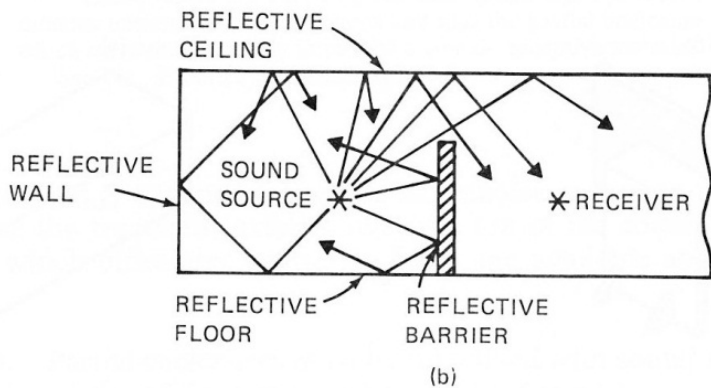
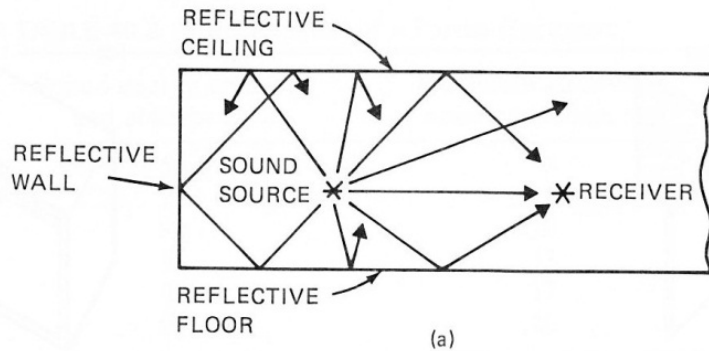


CORRECT

- Inertia slab with isolation at floor/ground
- Eliminate/isolated attachments at structures
- Flexible connections at pipes/ducts



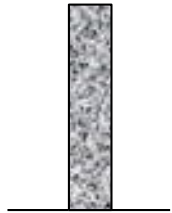
PARTIAL BARRIERS IN ROOMS



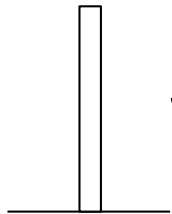
- Direct and reflected paths from source to receiver
- Reflective barrier blocks direct sound, but performance is limited by reflection and diffraction around barrier
- **MOST EFFECTIVE:**
Reflective barrier with use of sound absorption to minimize reflection at ceiling, walls, and at barrier

PARTIAL BARRIERS IN ROOMS

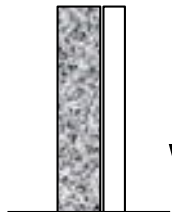
Example of Partial Barrier Options in a Room



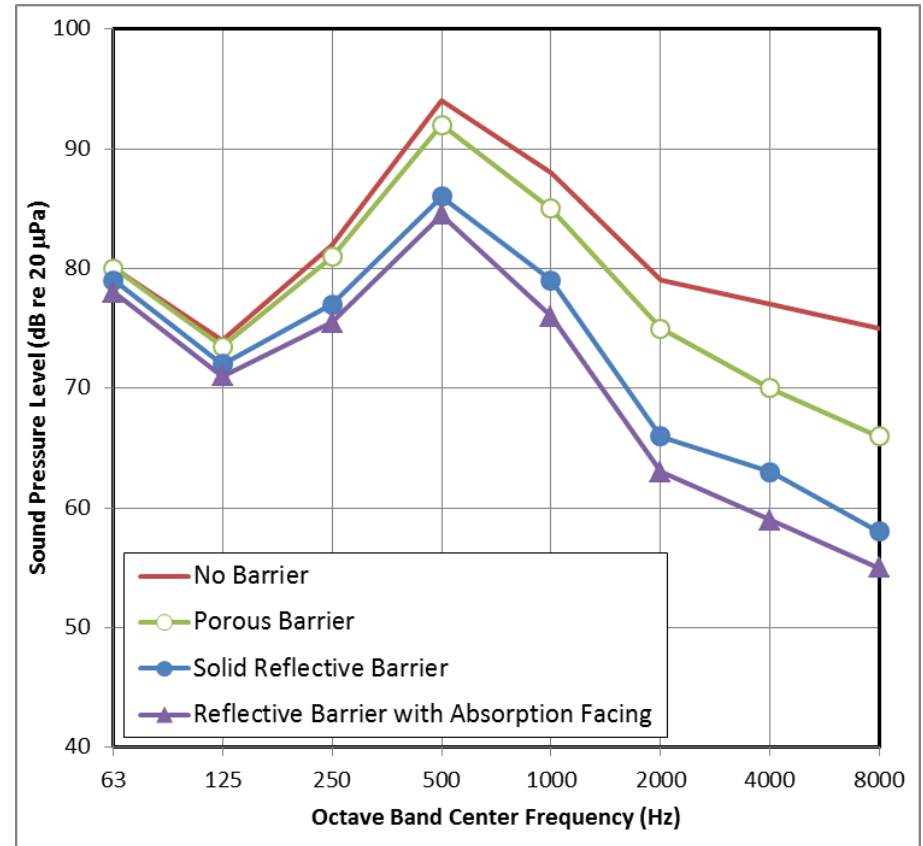
Porous
Barrier



Solid Reflective
Barrier



Reflective Barrier
With Absorptive Facing

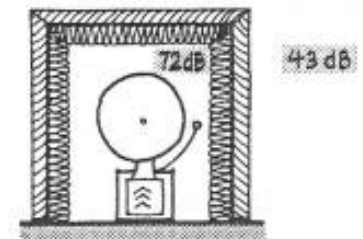
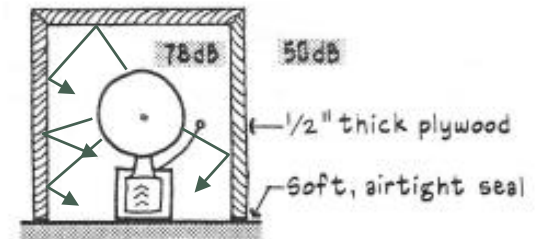
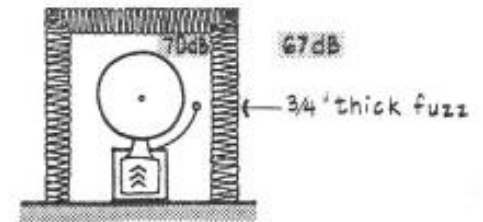
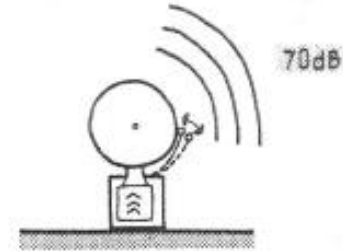


Complete enclosure around source

- Combination of a **sound blocking shell and sound absorbing liner** should be used
- Carefully select your dimensions such that the wavelengths produced by the source are not duplicated by the enclosure dimensions or integer multiples of the wavelength
 - *A match could amplify the sound, especially if no absorption is used!*
- Be concerned with heat build-up if applicable – may have to vent with acoustically treated duct

Reflective enclosure causes build-up of sound and raises noise level around source

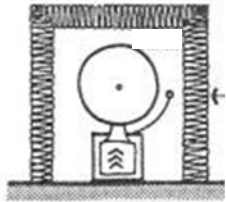
Absorptive lining minimizes build-up of sound and reduce noise levels in enclosure



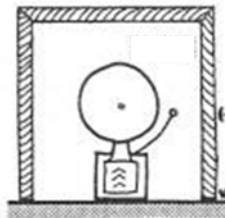
COMPLETE ENCLOSURE AROUND SOURCE



No Enclosure



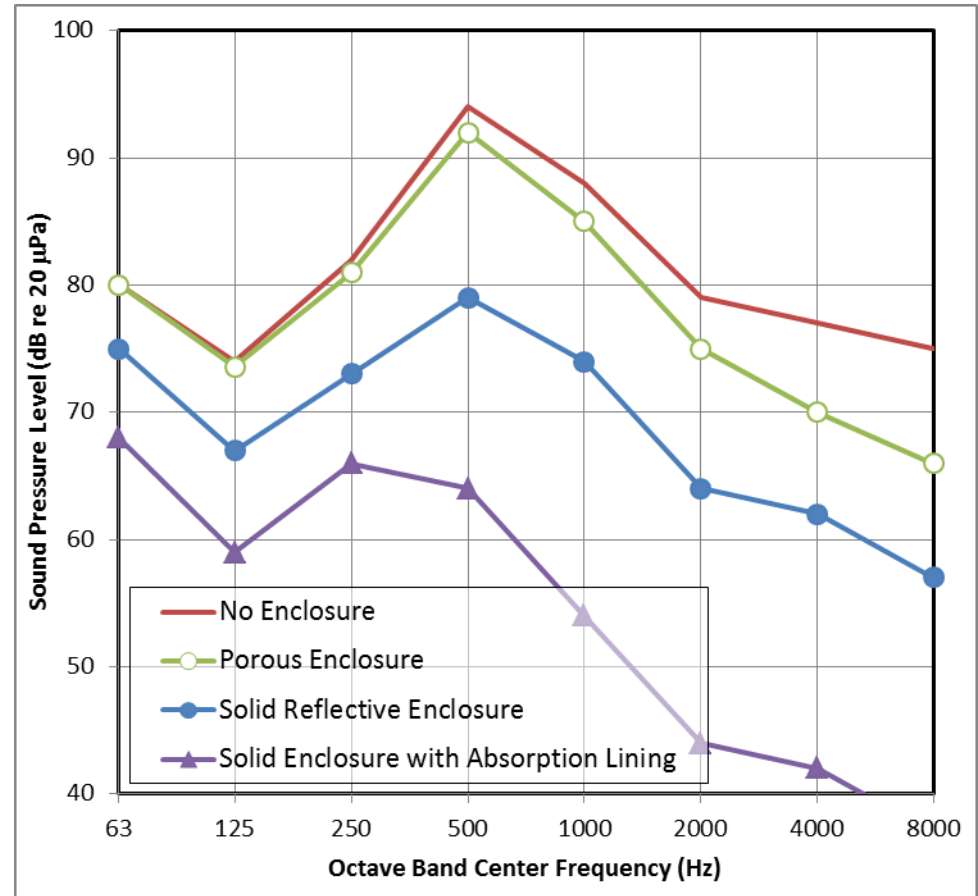
Porous Enclosure



Solid Reflective Enclosure



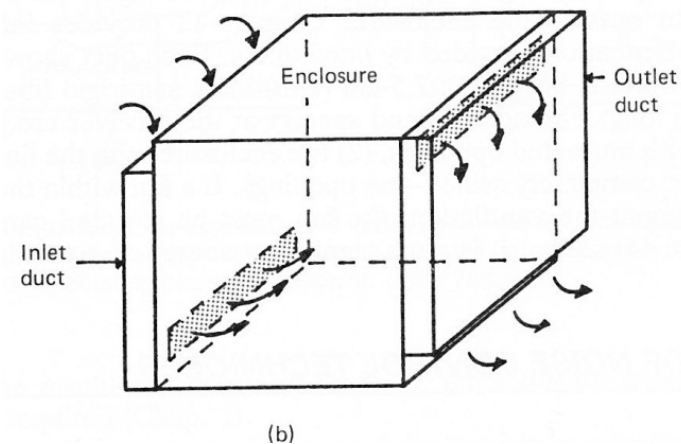
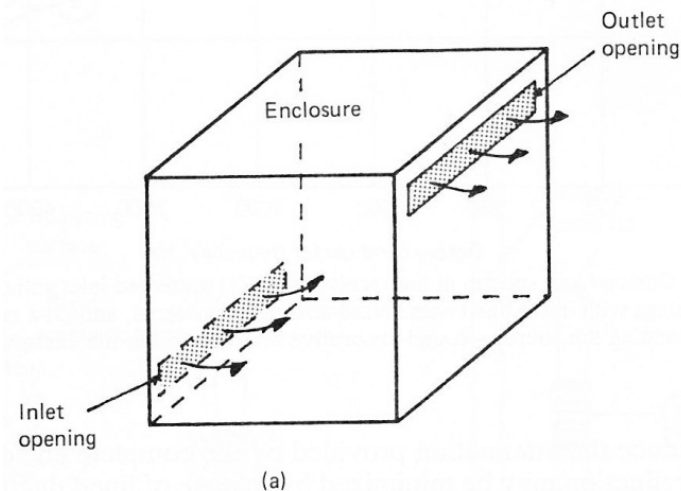
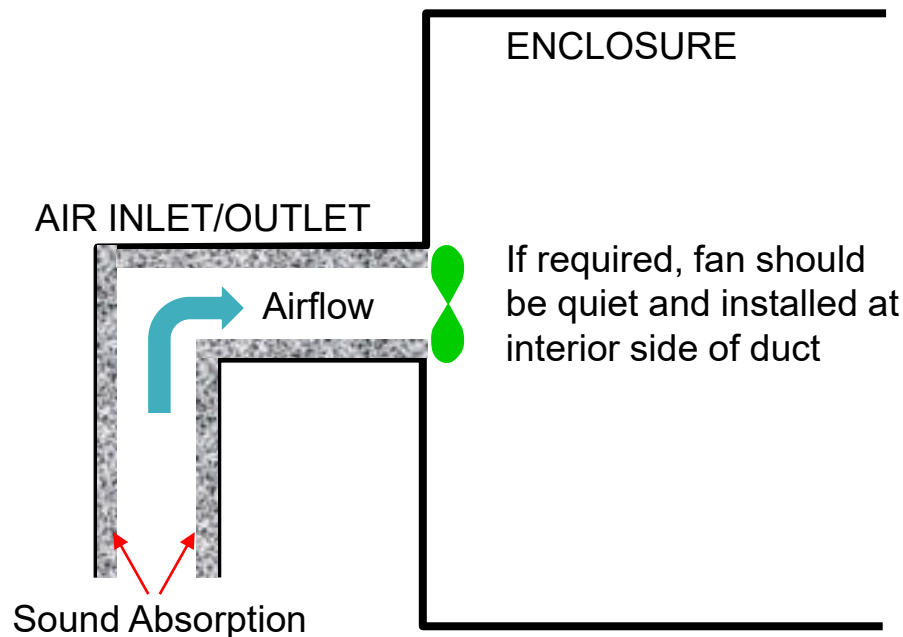
Solid Reflective Enclosure with Sound Absorption Lining



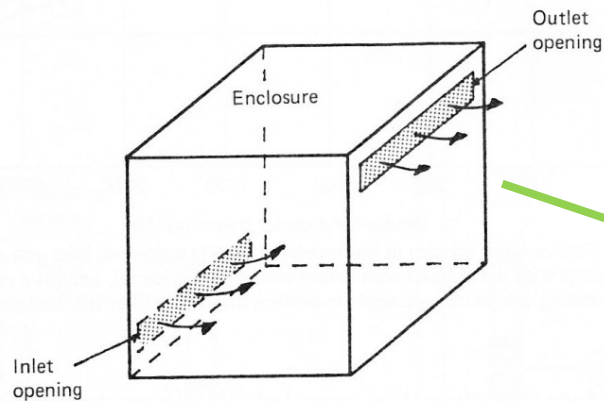
Example of Full Enclosure Options

VENTILATION FOR ENCLOSURES

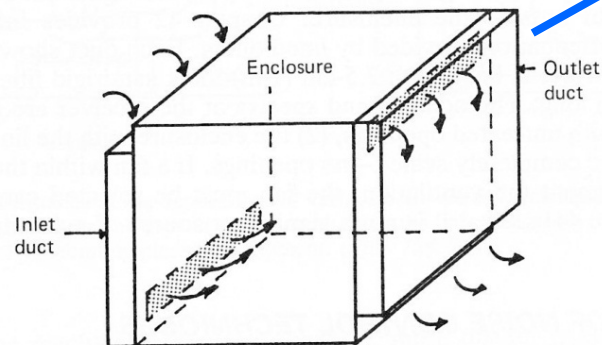
- Complete enclosures around noise sources may **require ventilation to prevent overheating of equipment**
- Openings will degrade barrier performance*
- Creating a torturous path for sound yet permitting airflow is the best solution
- Ideal inlets and outlets are long acoustically lined ducts/paths with at least one 90° elbow



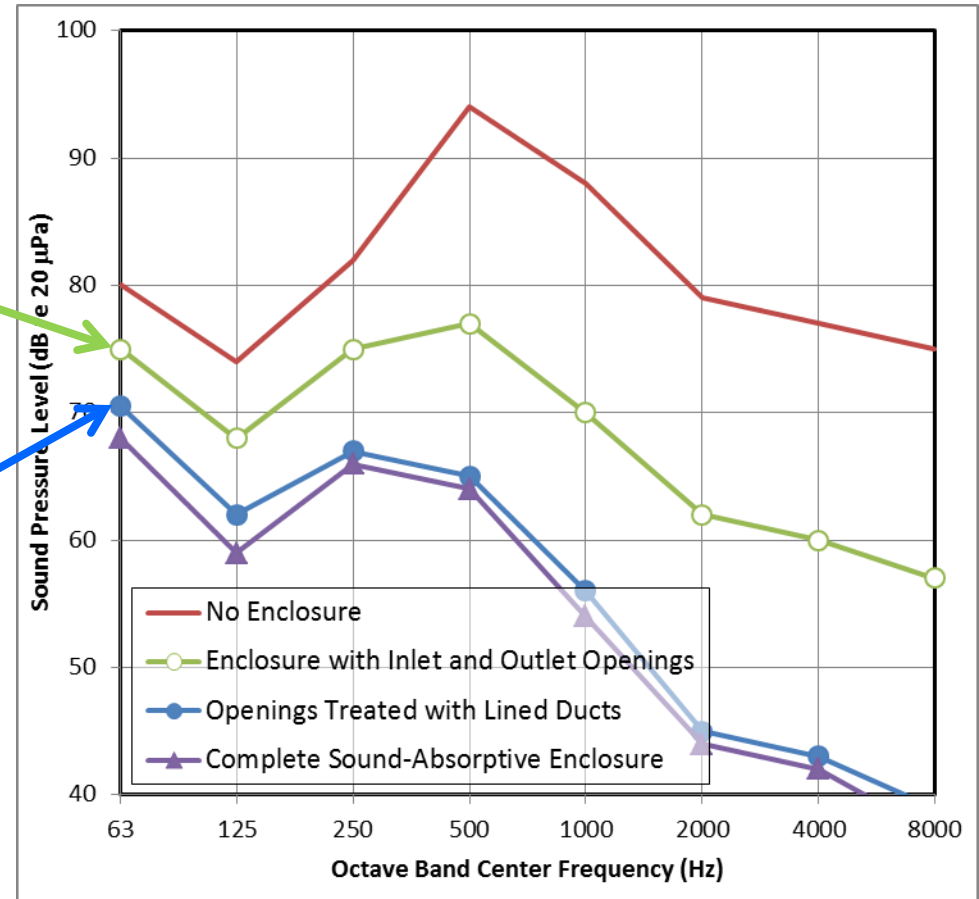
VENTILATION FOR ENCLOSURES



(a)



(b)



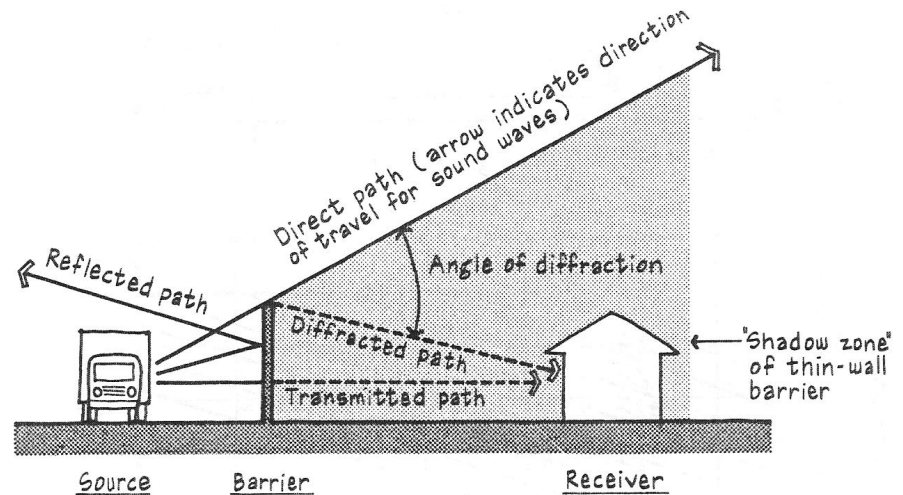
Example of Full Ventilation Options

ENCLOSURE CHECKLIST

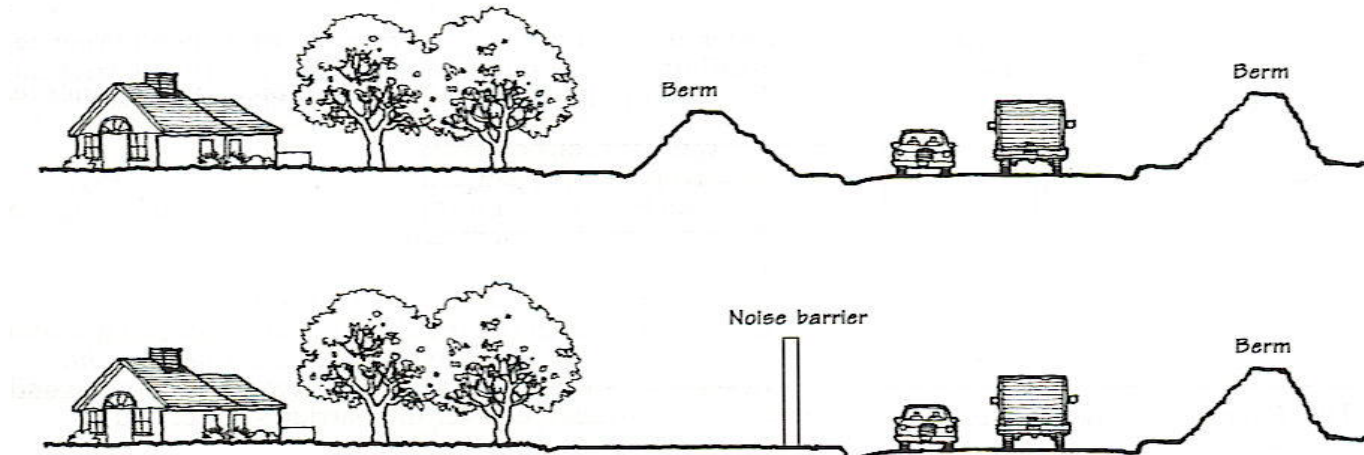
- Construct enclosure with dense, non-porous sides having high transmission loss (TL) or STC
- Mount vibrating source on massive inertia pad with vibration isolation
- Isolate or minimize contact of conduits, pipes, ductwork, etc., with walls of enclosure
- Use flexible connectors to reduce transmission of vibration of source
- Apply vibration damping material on enclosures, especially metal if structureborne noise is significant
- Eliminate all unnecessary openings, leaks and flanking paths through the enclosure (bottom, junction of sides, etc.)
- Seal around penetrations through the enclosure walls with flexible non-hardening caulk
- All doors and access panels should be dense and solid with air-tight gasketing
- Acoustically line as much of the interior surfaces of the enclosure with an effective sound absorption material
 - Consider absorption coefficient or NRC
 - Select material suitable for application – e.g., fire-rating, heat, moisture, etc.
 - Line or encapsulate material in wet or oily environments
 - Protect material from physical contact using perforated panels having at least 50% openness
- Use adequate ventilation when required
 - Create torturous path for sound – e.g., lined duct with at least one 90° elbow
 - If forced ventilation is necessary, select a quiet fan and install at interior side of duct

Outdoor noise Barriers

- Main goal: **break the line of sight**
- Diffraction over the barrier and around the ends of the barrier will limit its performance
 - To minimize end diffraction, the barrier length should be at least $4R$ where R is the shorter of the source-to-barrier or receiver-to-barrier distances
- Barrier materials: concrete, masonry, wood, metal, plastic
 - ~STC 33-35 minimum
- **Not thin plantings!!**
 - (evergreens >20 ft wide band block ~2-4 dB, 100 ft band of dense vegetation ~ 10 dB)
 - Although, there is some psychological value

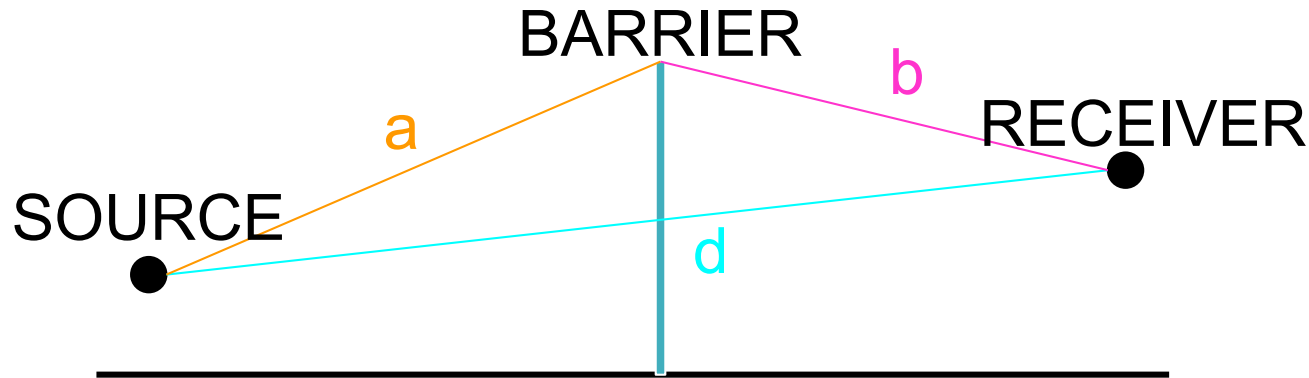


Outdoor noise Barriers



- Barriers such as **earth berms** or **constructed walls** can assist controlling exterior noise
- Care must be taken in their design, or else dB levels may be actually increased in some areas behind them
 - *Keep tree heights below barrier wall height to minimizing sound scatter*
 - **Do not plant trees on top of berms**

Outdoor noise Barriers



Main factors affecting barrier performance:

- Frequency
- **Path-Length-Difference, δ** : the difference between the path over the barrier versus the direct path through the barrier
 - i.e., $(a + b) - d$
 - The larger the δ , the greater the attenuation
- Sufficient TL of the barrier

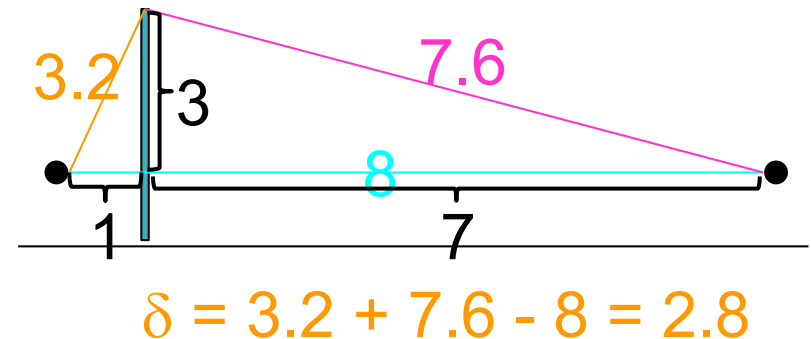
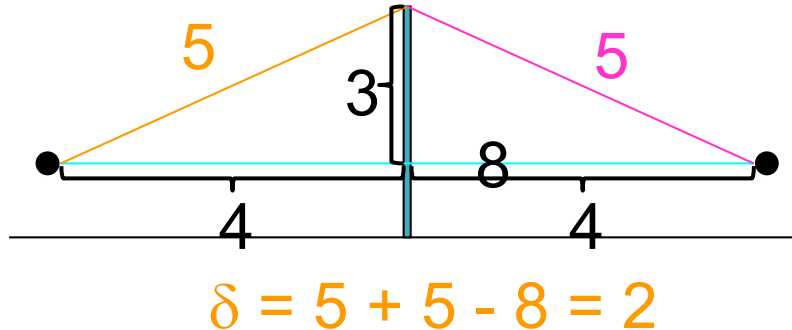
Outdoor noise Barriers

■ Frequency:

- Low frequencies (i.e., large wavelengths) diffract more around barriers and therefore are attenuated less

■ Path-Length-Difference:

- Locate the barrier close to the source or receiver to create the largest path-length-difference, δ (i.e., greater diffraction angle)
- *The worst location of a barrier is midway between the source and receiver.* Example:



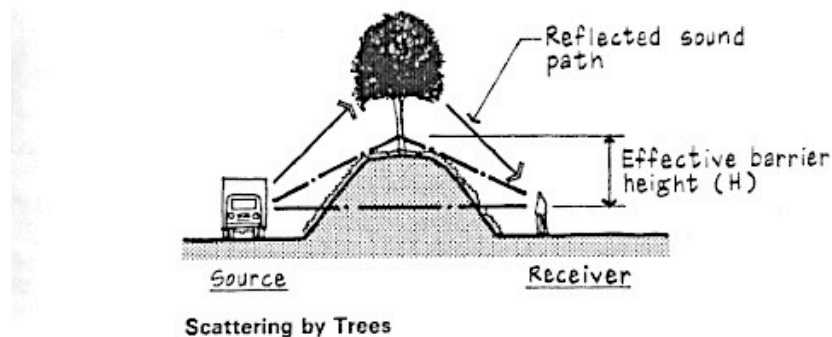
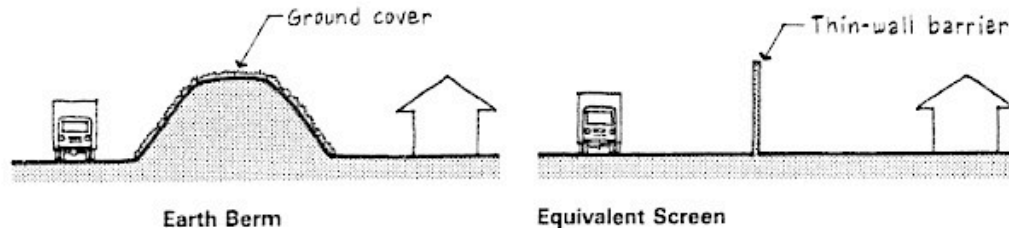
■ Barrier TL:

- Barrier TL should be greater than the barrier attenuation by at least 10 dB
- **Avoid** gaps, leaks and openings (e.g., backyard wooden privacy fence has negligible TL, therefore is not a noise barrier)

Outdoor noise Barriers

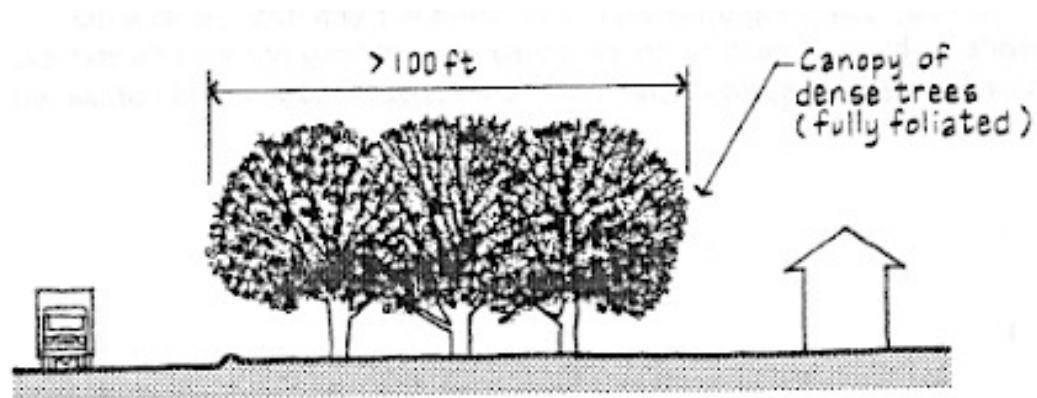
Earth berms

- Effectiveness reduced by reflective top surfaces: asphalt or concrete bicycle paths & walkways, deciduous trees
- Trees can reflect sound back towards receiver



Outdoor noise Barriers

- Intensity Level(IL) from vegetation
 - Single row of trees has no IL value – **NONE!**
 - 20 ft wide band of evergreens: ~2-4 dB IL
 - 100 ft wide band of dense plantings with closed canopy: ~10 dB



Outdoor noise Barriers

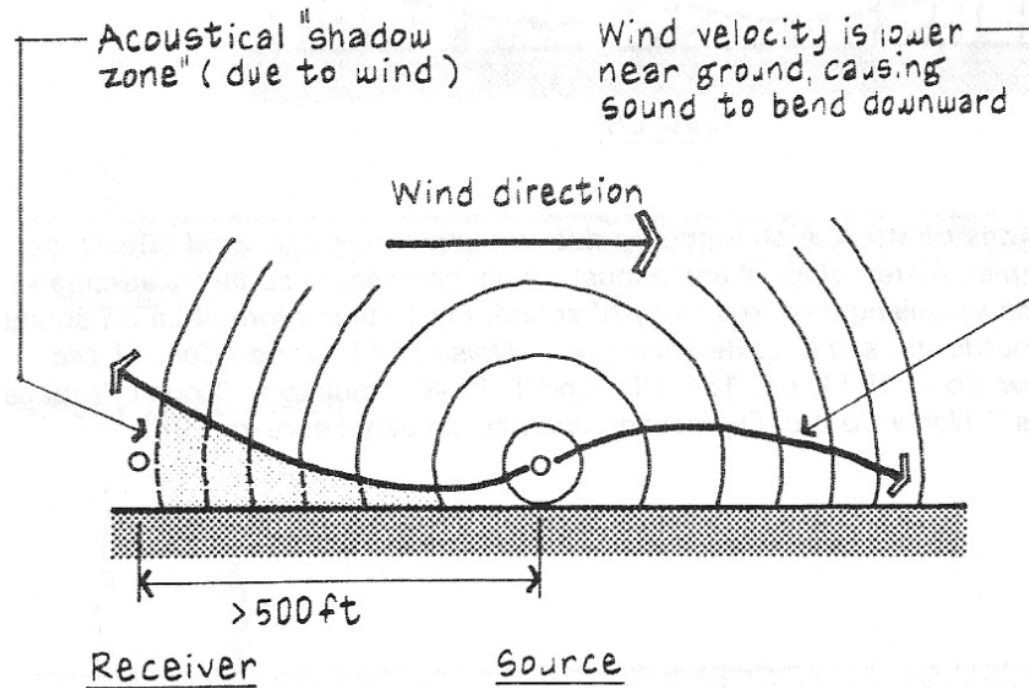
Other Considerations:

- Theoretical limitation for barrier attenuation is approx. 24 dB
- **Practical limitations make attenuation considerably less:**
 - Reflections from other nearby surfaces (buildings, equipment, etc.) can circumvent barrier
 - Gap at bottom of barrier for drainage or durability
 - Louvers in barrier for adequate ventilation and wind considerations
 - Acoustic louvers can be used
- **Sound absorption material** is often applied to source side of barrier to minimize sound build-up and reduce pressure levels at source
 - Must be weatherproof if exposed to elements
- **Barrier width should be 5 times wider than height to minimize end diffraction**
- Use well-sealed access panels for barriers close to equipment requiring regular servicing
- Robust barrier to withstand wind loading
- Can use 2-sided or 3-sided barriers for better shielding and strength

OUTDOOR SOUND - Wind Effects

Typical effects from wind

- Upwind: sound is bent upward, attenuating the sound
- Downwind: sound is bent downward, increasing the sound
- Example at 10 mph: +/- 10 dB at distances greater than 500 ft



OUTDOOR SOUND - Temperature Effects

Typical effects from temperature

■ Daytime

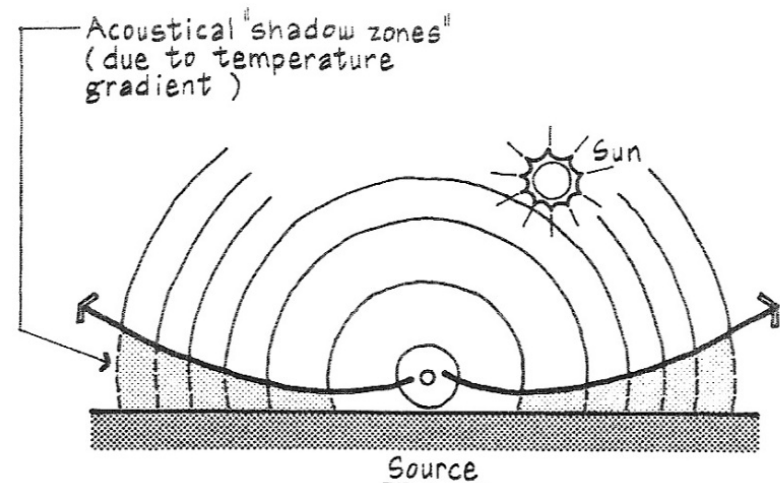
- Air temperature usually decreases with increasing height above ground (temperature lapse)
- **Sound bends upward** (depicted in sketch)

■ Nighttime

- Air temperature usually decreases with decreasing height above ground due to radiation cooling of ground (temperature inversion)
- To height of 100 meters or more

- **Sound bends downward**

- Difference can be 10 dB at 1000 ft or more



Week 13

Noise Control, Speech Intelligibility

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Speech Intelligibility

- For speech/conversation to be heard clearly in a space, the reverberation time is important. A **low reverberation time is essential**, and any reflections should reinforce speech without creating echoes
- Another important factor for speech is the level of background (i.e., ambient) noise in the space. High levels of ambient noise (i.e., drown-out) mask speech and reduce the percentage of speech heard clearly

Speech Intelligibility

- *Speech Intelligibility* (SI) can be computed during design or measured after construction
- SI is a function of the difference between the sound level of speech (the signal) and the background level of sound (the noise)
 - Extraneous noise tends to mask (cover) sound, which reduces SI
 - *Signal-to-noise ratio* (S-N ratio) is the difference of signal SPL and the background sound SPL, expressed in dB
 - Octave bands from 250 Hz to 4 kHz are most important for speech

Articulation Index

- **Articulation Index (AI)** is a calculation of level of **speech intelligibility**
- Function of RT (Reverberation time), reflecting surfaces, TL, echoes, and background sound level
- AI runs from 0.0 (no speech intelligibility and perfect speech privacy) to 1.0 (perfect speech intelligibility and no speech privacy)
 - High intelligibility/low speech privacy – lecture hall, classrooms, drama theater, cinema, courtroom
 - Low intelligibility/high speech privacy – through private office walls, open plan offices

Articulation Index

- This table subjectively describes the speech intelligibility in a room
- For “good intelligibility”, an AI of 0.5 is recommended
- If a variety of talkers and listeners are using the space (typical), a minimum of AI of 0.7 is recommended

Table 14.2 Relationship Between AI and Speech Intelligibility

AI	Speech intelligibility
≥ 0.7	Very good
0.5 - 0.7	Good
0.3 - .5	Marginal
< 0.3	Poor

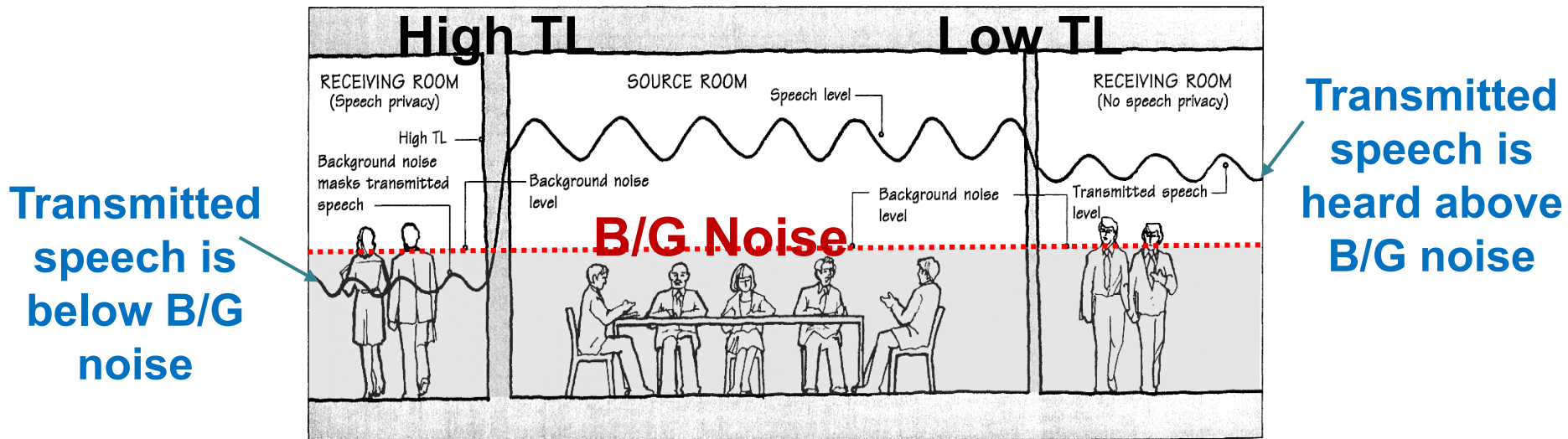
Speech Privacy

- Design goal for private offices, medical exam rooms, counselors offices, jury deliberation rooms, etc., is to provide a high degree of speech privacy
- Low AI implies high speech privacy
- A low SI or AI between adjacent private rooms or adjacent cubicles (open plan) is desirable

Degree of Privacy	Articulation Index (AI)
Confidential	< 0.05
Normal (freedom from distraction)	0.05 to 0.20
Marginal or poor (good for eavesdroppers)	0.20 to 0.30
Poor or none	> 0.30

Speech Privacy

- Speech privacy between adjacent spaces can be improved in two ways, incorporated separately or in combination
 1. Improve transmission loss of the partition
 2. Increase the background noise level (HVAC or sound masking system) in the adjacent space
- *Thus, if the environment is made quieter, the TL of the partition must be increased for the same speech privacy level. Likewise, if the background masking sound is increased, the partition TL can be decreased.*



14.5 Effect of partition's transmission loss and background noise level on speech privacy.

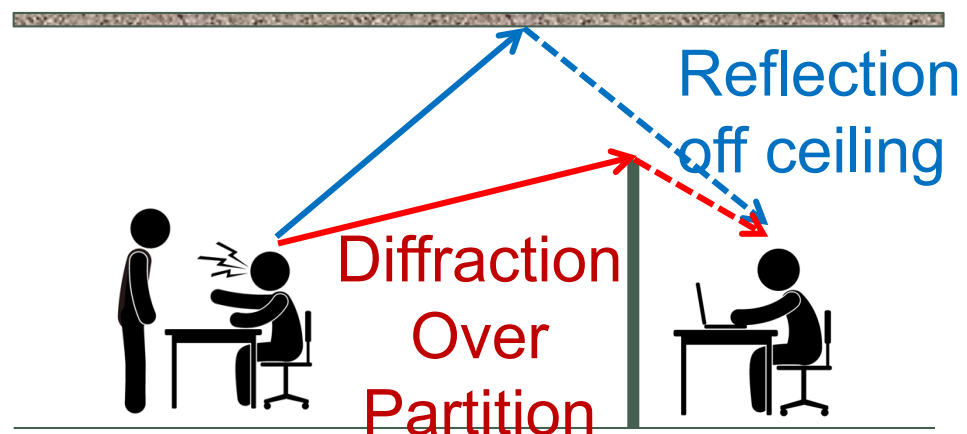
Speech Privacy *in Open Plan Offices*

- *Speech privacy is one of the most important design issues in an open plan office*
- An open plan office is a large room where individual workstations are separated by partial height partitions
- Achieving speech privacy is far more difficult for open plan offices than “conventional” (full height partition) offices
- Reasonable levels of speech privacy are obtainable if the following design guidelines are followed

Speech Privacy *in Open Plan Offices*

Ceiling

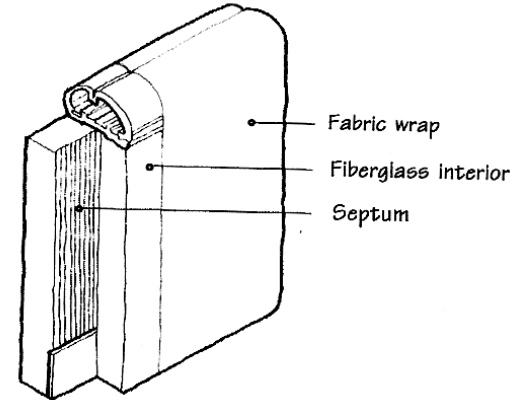
- The ceiling is the *most important acoustical element* in open plan offices
- Must be highly absorptive: NRC of 0.75 or higher
- Minimize ceiling mounted light fixtures (good sound reflectors) by providing workstation lighting and/or indirect ceiling lighting



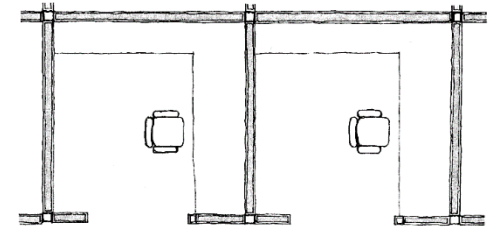
Speech Privacy *in Open Plan Offices*

Screens

- Partial height partitions (screens) should have two functions:
 - **Reduce direct sound** between workstations
 - Screens should be at least 5 ft tall and 8 ft long
 - **Absorb incident sound**
- Typical construction:
 - Sound insulating septum: 1/8" plywood or hardboard, or heavy aluminum foil
 - Sound absorber: 1" to 1-1/2" thick fiberglass laminated to septum on both sides (In practice, a tack board surface is frequently used which does not perform as well as fiberglass.)
- *Where a workstation is close to a wall or column, these surfaces should be covered with a similar absorptive material.* (Unfortunately, this is rarely seen in practice.)



Speech Privacy *in Open Plan Offices*



Workstation Distance and Orientation

- Orient workers such that they face in the same direction since speech is directional. *Avoid face-to-face orientation*
- Alternately, orient workers *toward a sound absorbing surface*
- Speaker phones should be avoided

Floors

- *Carpet* should be installed to reduce impact and chair/roller noise.

Windows

- *Locate workstations away from reflective windows.* Use area next to windows as circulation corridors.

Speech Privacy *in Open Plan Offices*

Sound Masking System (Artificial B/G Noise)

- *A continuous and constant background broadband (many frequencies) sound will mask sounds at similar frequencies*
- While HVAC noise does mask sound, its consistency cannot be relied upon. Therefore, *sound masking systems* are frequently introduced into open plan office spaces.
 - A sound masking system consists of a distributed speaker system that plays a constant broadband spectrum which serves as masking sound.

Sound Masking Systems

- A sound masking system is typically designed by **the manufacturer or an acoustical consultant**
- Masking systems can be used in other types of spaces where speech privacy is important.
 - Courtroom sound masking systems can be turned on by the judge when a conference which the jury must not hear is taking place at the bench
 - Medical care facilities in areas where patient privacy is important. This may become more common in the future with HIPAA (Health Insurance Portability and Accountability Act of 1996) regulations.

Week 13

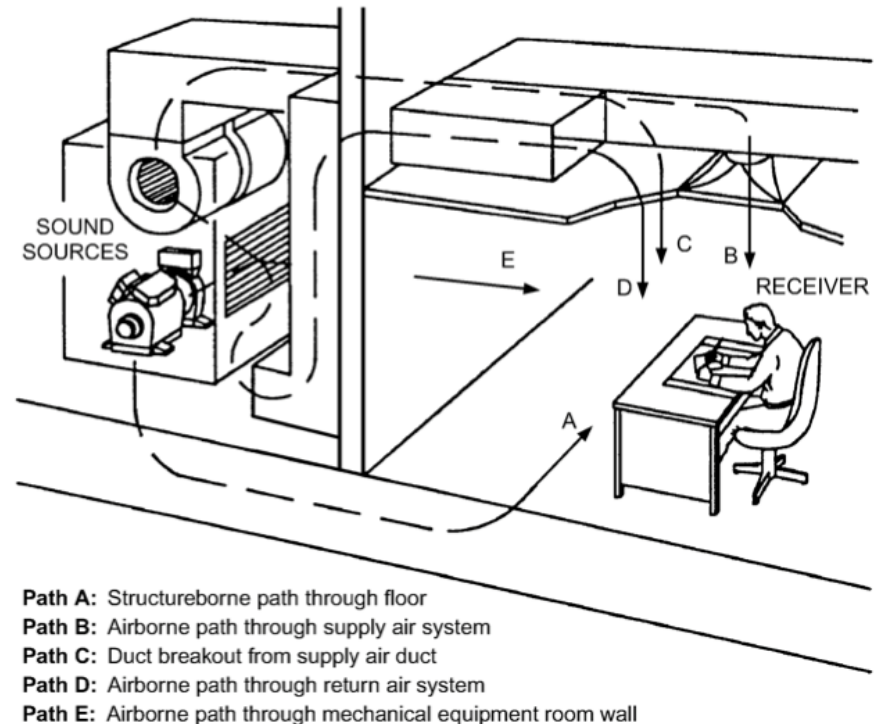
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Mechanical Noise Control

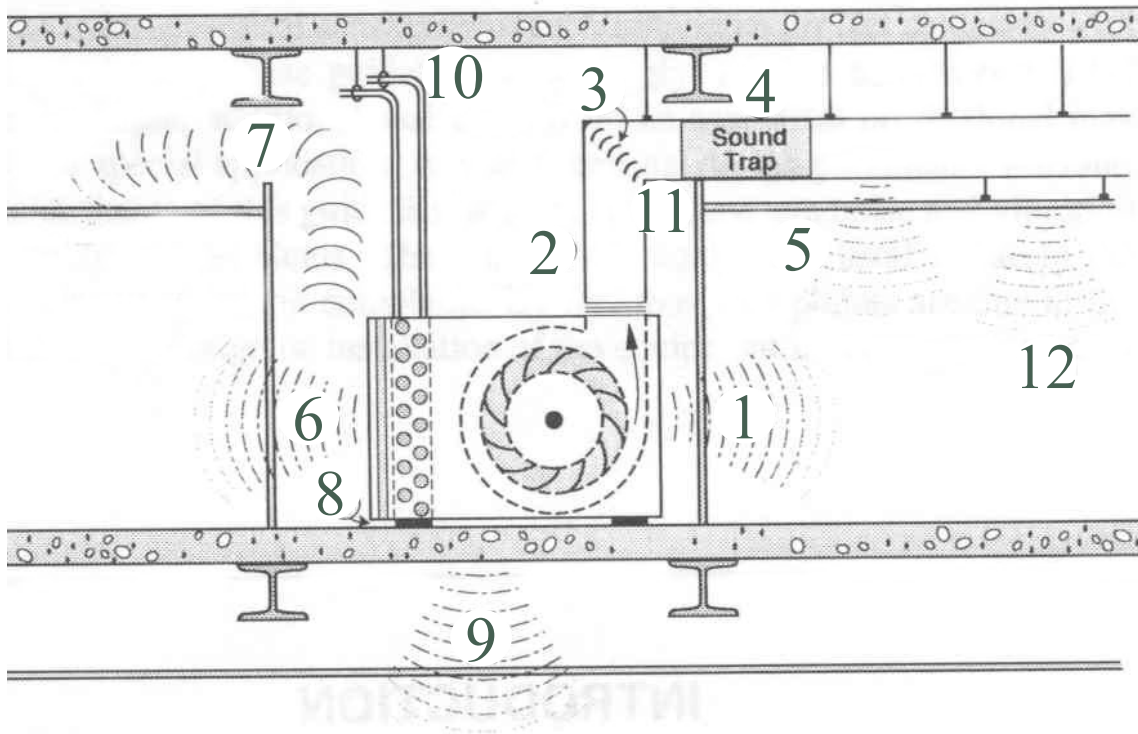
Source – Path – Receiver

- Source: AHU, fans, terminal units, fan coil units, unit ventilators
 - Paths: ductborne, radiated, structure-borne
- Source: Diffusers, grilles, registers
 - Path: radiated
- Source: Chillers, cooling towers, pumps, dry coolers
 - Paths: piping radiated, equipment radiated, structure-borne
- Source: Airflow turbulence in ductwork (i.e., path)
 - Paths: ductborne, radiated



MECHANICAL NOISE CONTROL

Sound Paths for HVAC Systems



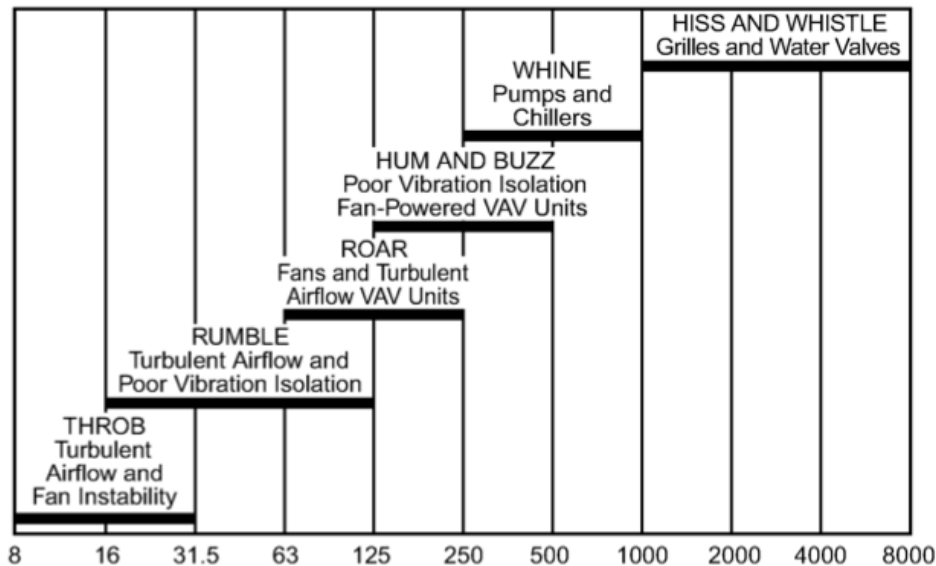
1. Coupling with wall
2. Poor flow condition
3. Turning vanes
4. Silencer location
5. Breakout noise
6. Too close to wall
7. No return silencer
8. Inadequate isolation
9. Structural stiffness
10. Rigid hangers
11. Duct touching drywall
12. Ceiling hung from ductwork supports

Mechanical Noise Control

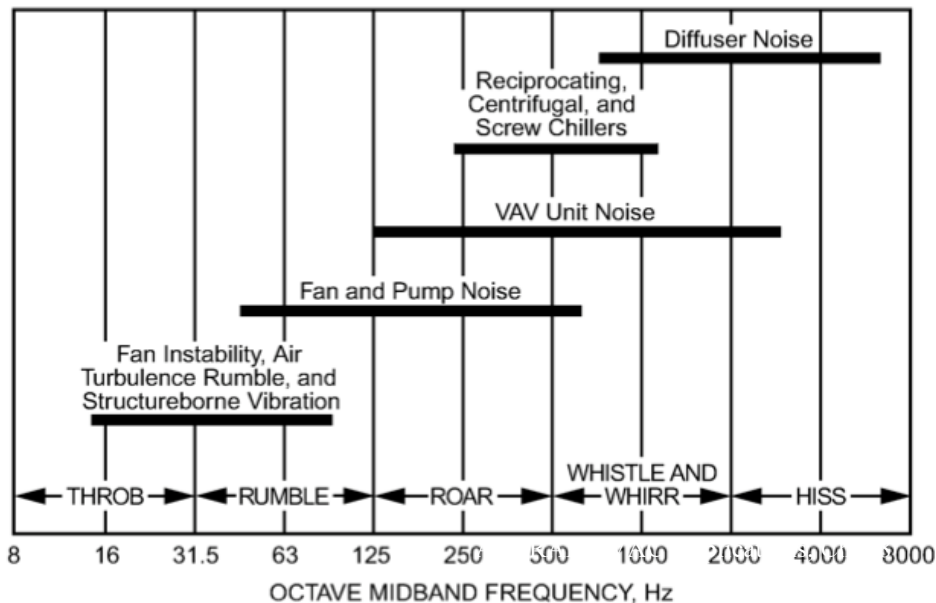
Noise annoyance is generally:

- Proportional to loudness
- Greater for high frequency than low
- Greater for intermittent noise than continuous
- Greater for pure tone than broadband
- Greater for moving or unlocatable (reverberant) noise than for fixed-location
- Greater for “information bearing” than “nonsense” noise

Mechanical Noise



- Frequency ranges of likely sources of sound-related complaints



- Frequencies at which different types of mechanical equipment are generally dominant

HVAC Criteria: dBA & dBC

- dBA & dBC

- Straightforward, easy to measure
- Correlate with judgment of sound level
- Shortcomings: no quality or spectral balance information
- example: Used in ANSI S12.60 for background levels in schools (standard is referenced in LEED for Schools) for simplicity

Learning space ^{a)}	Greatest one-hour average A- and C-weighted sound level of exterior-source background noise _{b), f)} (dB)	Greatest one-hour average A- and C-weighted sound level of interior-source background noise _{c), f)} (dB)	Maximum permitted reverberation times for sound pressure levels in octave bands with midband frequencies of 500, 1000, and 2000 Hz (s)
Core learning space with enclosed volume $\leq 283 \text{ m}^3$ ($\leq 10\,000 \text{ ft}^3$)	35 / 55	35 / 55	0.6 s ^{e)}
Core learning space with enclosed volume $> 283 \text{ m}^3$ and $\leq 566 \text{ m}^3$ ($> 10\,000 \text{ ft}^3$ and $\leq 20\,000 \text{ ft}^3$)	35 / 55	35 / 55	0.7 s
Core learning spaces with enclosed volumes $> 566 \text{ m}^3$ ($> 20\,000 \text{ ft}^3$) and all ancillary learning spaces	40 / 60 ^{d)}	40 / 60 ^{d)}	No requirement
a) See 3.1.1.1 and 3.1.1.2 for definitions of core and ancillary learning spaces. b) The greatest one-hour average A- and C-weighted interior-source and the greatest one-hour average A- and C-weighted exterior-source background noise levels are evaluated independently and will normally occur at different locations in the room and at different times of day. c) See 5.2.2 for other limits on interior-source background noise level. d) See 5.2.3 for limits in corridors adjacent to classrooms. e) See 5.3.2 for requirement that core learning spaces $\leq 283 \text{ m}^3$ ($\leq 10\,000 \text{ ft}^3$) shall be readily adaptable to allow reduction in reverberation time to 0.3 s. f) The design location shall be at a height of 1 m above the floor and no closer than 1 m from a wall, window, or fixed object such as HVAC equipment or supply or return opening. See A.1.3 for measurement location.			

ANSI S12.60 *Acoustical Performance Criteria, Design Requirements, and Guidelines for Schools, Part 1: Permanent Schools*

HVAC Criteria: ASHRAE

HVAC noise should:

- Have balanced contribution over the whole spectrum
- No tones
- No fluctuations

Table 1 Design Guidelines for HVAC-Related Background Sound in Rooms

Room Types		Octave Band Analysis ^a	Approximate Overall Sound Pressure Level ^a	
		NC/RC ^b	dBA ^c	dB(C) ^c
Rooms with Intrusion from Outdoor Noise Sources ^d	Traffic noise	N/A	45	70
	Aircraft flyovers	N/A	45	70
Residences, Apartments, Condominiums	Living areas	30	35	60
	Bathrooms, kitchens, utility rooms	35	40	60
Hotels/Motels	Individual rooms or suites	30	35	60
	Meeting/banquet rooms	30	35	60
	Corridors and lobbies	40	45	65
	Service/support areas	40	45	65
Office Buildings	Executive and private offices	30	35	60
	Conference rooms	30	35	60
	Teleconference rooms	25	30	55
	Open-plan offices	40	45	65
	Corridors and lobbies	40	45	65
Courtrooms	Unamplified speech	30	35	60
	Amplified speech	35	40	60
Performing Arts Spaces	Drama theaters, concert and recital halls	20	25	50
	Music teaching studios	25	30	55
	Music practice rooms	30	35	60
Hospitals and Clinics	Patient rooms	30	35	60
	Wards	35	40	60
	Operating and procedure rooms	35	40	60
	Corridors and lobbies	40	45	65
Laboratories	Testing/research with minimal speech communication	50	55	75
	Extensive phone use and speech communication	45	50	70
	Group teaching	35	40	60
Churches, Mosques, Synagogues	General assembly with critical music programs ^e	25	30	55
Schools ^f	Classrooms	30	35	60
	Large lecture rooms with speech amplification	30	35	60
	Large lecture rooms without speech amplification	25	30	55
Libraries		30	35	60
Indoor Stadiums, Gymnasiums	Gymnasiums and natatoriums ^g	45	50	70
	Large-seating-capacity spaces with speech amplification ^g	50	55	75

Plumbing Sound Criteria

Plumbing Noise

- Short duration or transient
- Examples: water supply, waste lines, jetted bathtubs
- ASHRAE criteria intended for multifamily, hospitals, educational facilities and office buildings.
- High-end luxury condominiums and private homes should be 5-10 dB lower than levels in table

Receiving (listening) room	L_{max} (slow response)
Residential bedroom/living room/dining room	35
Hospital patient room/classroom	40
Private office/conference room	40
Residential bathroom/kitchen	45
Open office/lobby/corridor	50

A-Weighted Level (dBA)

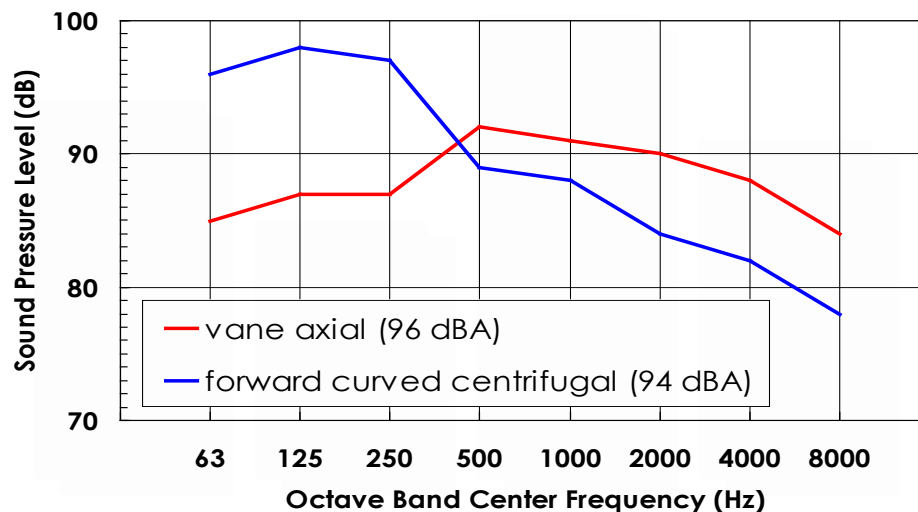
Equipment Noise sources

EQUIPMENT AS NOISE SOURCES

- Fans
- Chillers
- Cooling towers
- Pumps
- Terminal units (VAV, CAV)
- Fan coil units
- Unit ventilators

Fan Types & Selection

- Specify quiet equipment
- Larger/slower is generally quieter
- Efficiency & operating point – select for optimal operation
- Centrifugal, vane axial, mixed flow, propeller



Chillers & cooling towers

Chillers

- Low noise fan blades (can be airfoil, wide chord, more blades, swept blades)
- Multi stage or VFD (Variable Frequency Drive) fans
- Compressor blankets (enclosure wrap with a sound absorber and sound blocking mass loaded vinyl)

Cooling towers

- Low noise fan blades
- Two stage or VFD fans
- Water silencer

Pumps and terminal units

Equipment as sound sources

Pumps

- Airborne noise usually is not a problem, but structure-borne can be – will address in vibration lecture

Terminal units (VAV, CAV), Fan coil units, Unit ventilators

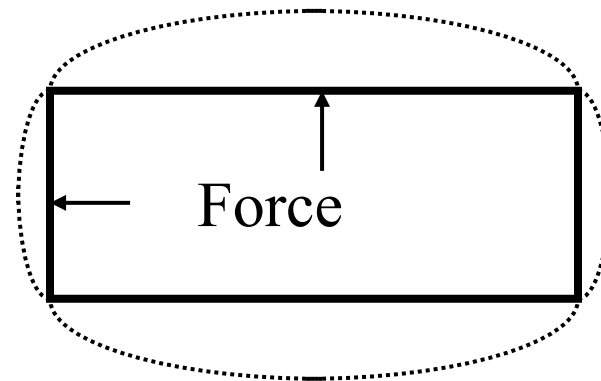
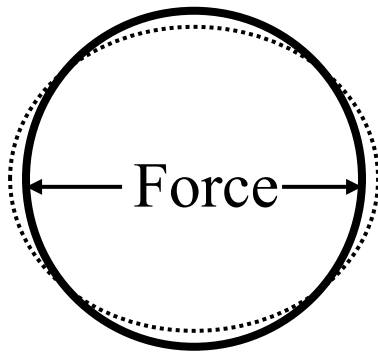
- Heavier gauge casing to decrease radiated
- Acoustically treated inlets and outlets
- Multi-stage fans

Air Delivery Noise Control Guidelines

- Ducts
- Fittings
- Terminal units
- Dampers
- End devices
- Wall & floor duct and piping penetrations

Ducts – breakout noise and rumble

- Rectangular duct is susceptible to breakout* & rumble noise if near the fan
- Circular ducts are less prone to breakout & rumble but will carry noise further up/downstream



* Breakout noise – noise in ductwork radiating into space outside of duct

Ducts

- Locate main ducts over corridors, bathrooms, storage rooms, etc.
- Slow air flow is quieter than fast air flow
 - *Keep velocities at or below ASHRAE recommended speeds for the space use*

Main Duct Location	Design RC	Max fpm rect. duct	Max fpm round duct
Above suspended acoustic ceiling	45	2500	4500
	35	1750	3000
	25	1200	2000

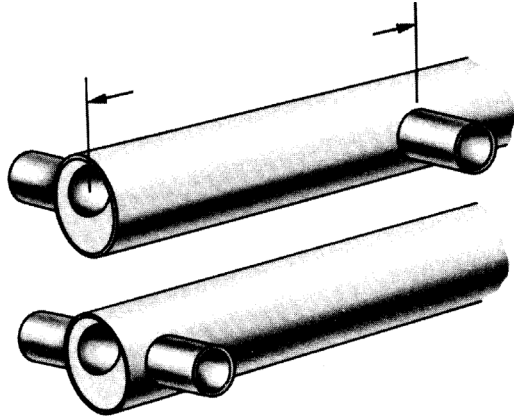
- Branch ducts at 80% of listed values & Final runouts at 50% or less of listed values
- Elbows and other fittings can increase airflow noise substantially

excerpt from 1999 ASHRAE Applications Handbook, Section 46, Table 3, p. 46.8

Ducts

Smooth flow is quieter than turbulent flow

- Keep 4 to 6 duct diameters between takeoffs to reduce turbulence noise



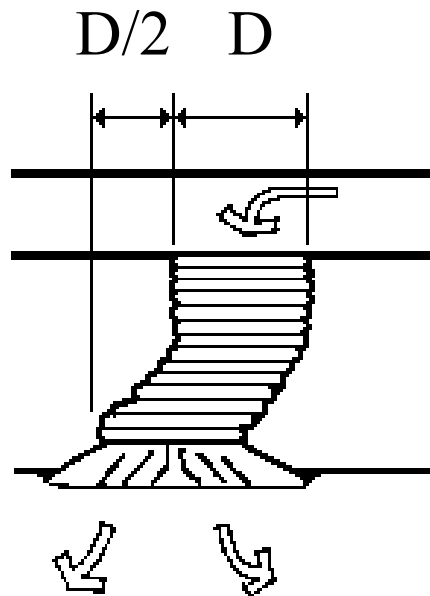
4-6 main duct diameters

Opposite connections

- If opposite connections must be used, *keep size & flow as equal as possible*

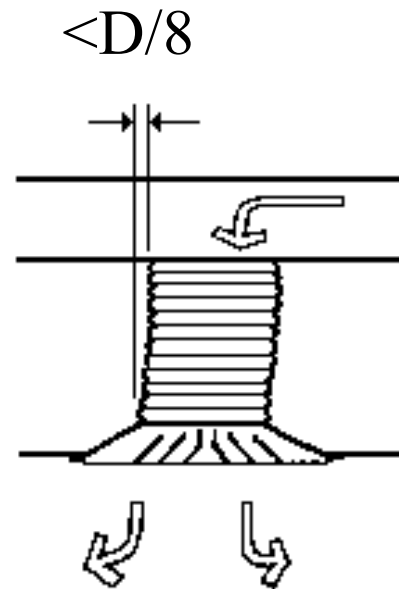
Ducts

- Flexible ducts are prone to breakout noise and turbulence noise



Poor alignment

(levels 12 to 15 dB higher than manufacturer's ratings)



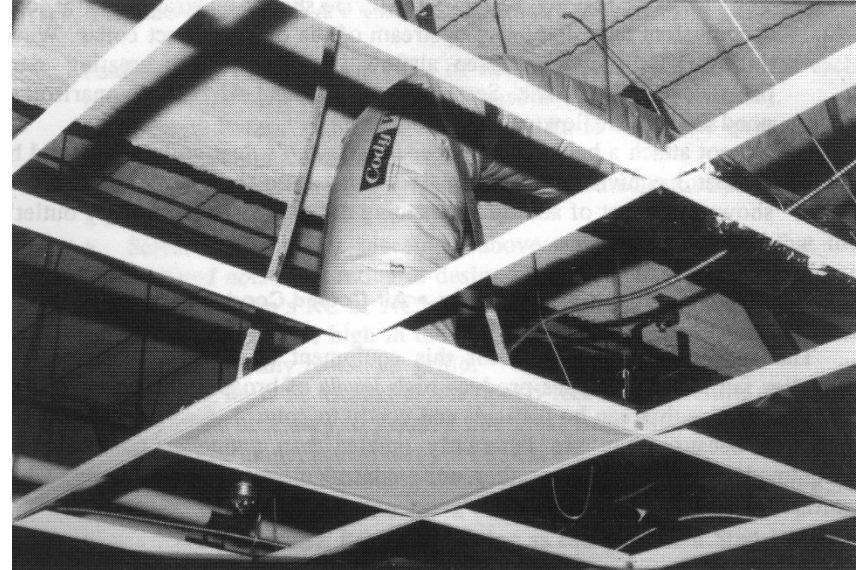
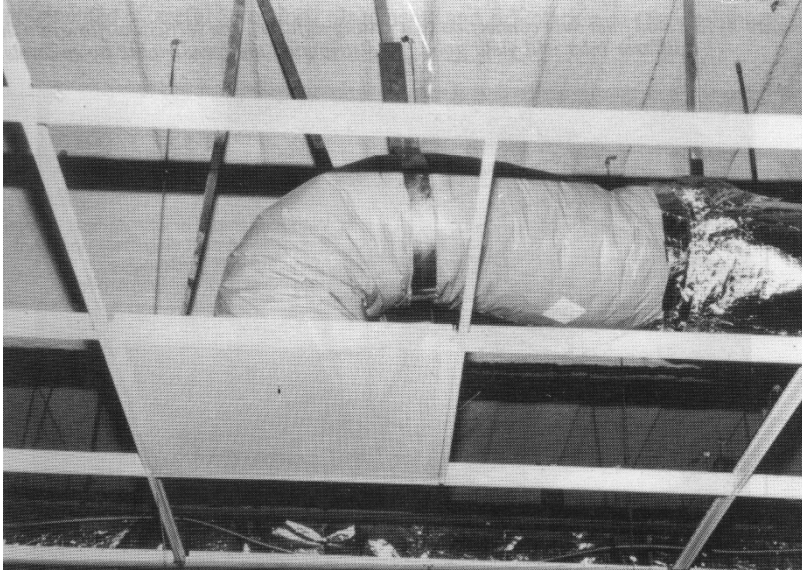
Good alignment

source: M. David Egan, *Architectural Acoustics*

Ducts

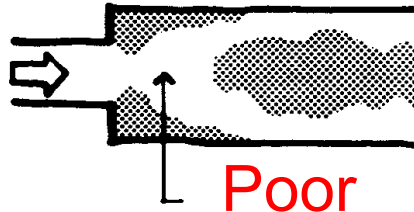
Ductwork should be straight for 3 or more duct diameters before air device

→ Otherwise, it can result in up to 15 dB higher than rating

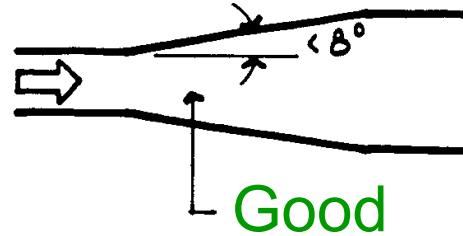


Fittings

■ Transitions

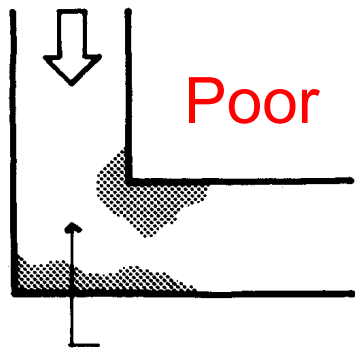


Large abrupt increase in duct size creates turbulence and noise

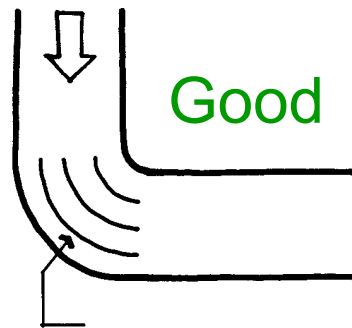


Smooth transition (provides gradual change in duct size)

■ Elbows



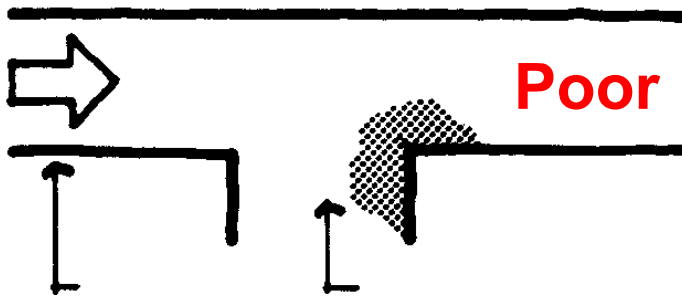
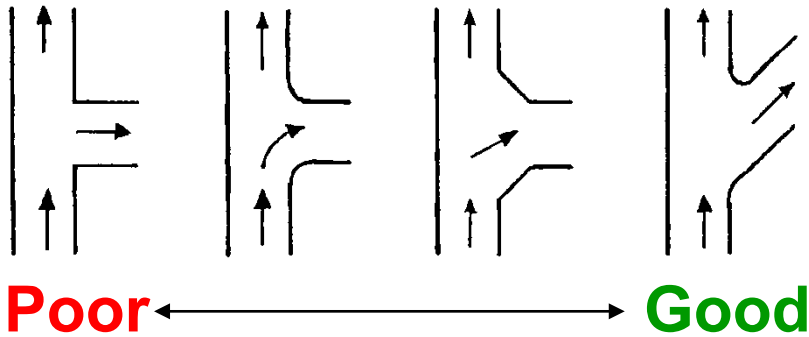
Turbulent airflow at sharp turns



Turning vanes (to smooth airflow & prevent air from "hugging" outside radius of elbow)

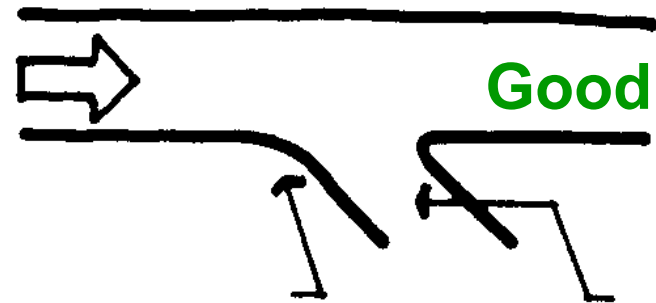
Fittings

▪ Branch/Takeoff



Main
air duct

Branch takeoff
(can be serious
generator of noise)

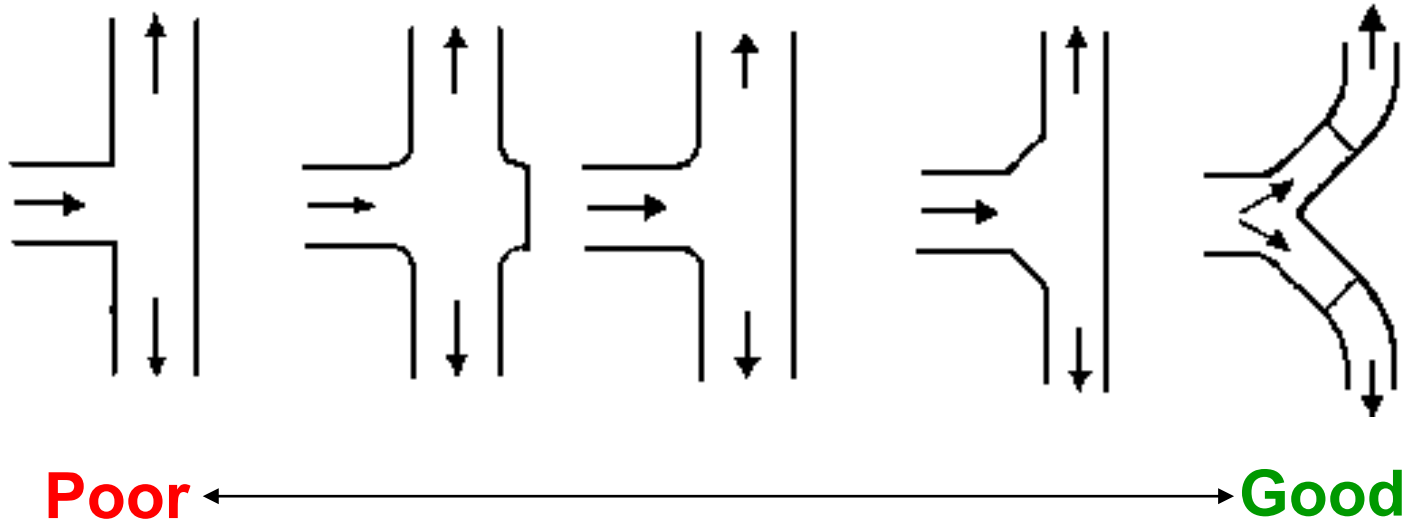


Long-radius
smooth turn

Angular
branch
takeoff

Fittings

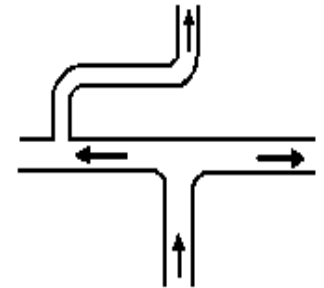
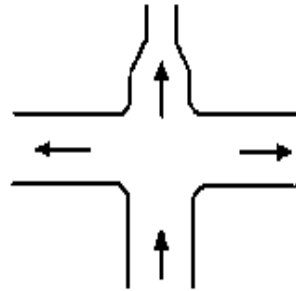
- Tee
with approximately equal flow



Fittings

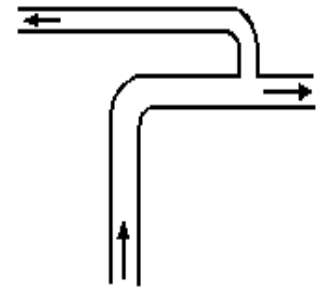
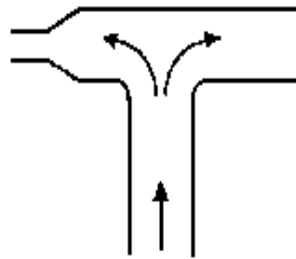
- Cross

with flow in one branch
much larger than the other



- Tee

with flow in one branch
much larger than the other



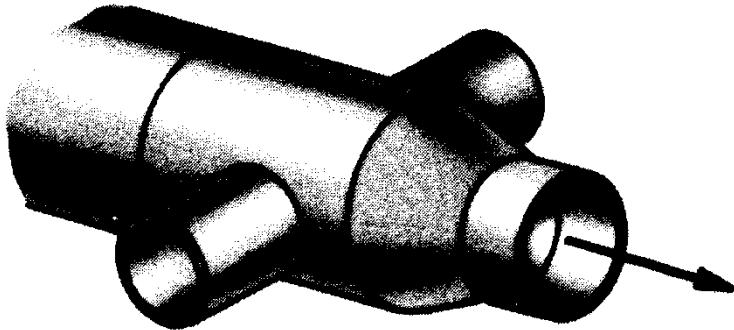
Poor ← → **Good**

Fittings

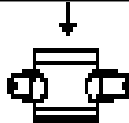
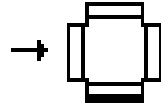
- Divided flow



- Three way flow



Minimum 33% of
total upstream flow

Fitting		Loss Coeff.
Y-branch plus 45° elbows		0.22
Capped cross with conical taps		4.5
Capped cross		4.5

Terminal Units

Try not to locate over spaces of RC 35 or less

(conference, classroom, OR, exam)

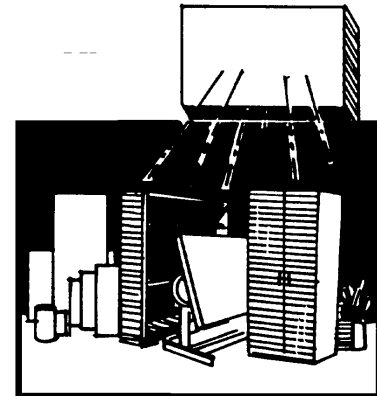
- If there is a choice, locate over room with greater RC (e.g., open plan instead of private office or corridor instead of library)



Private Office



Conference Room



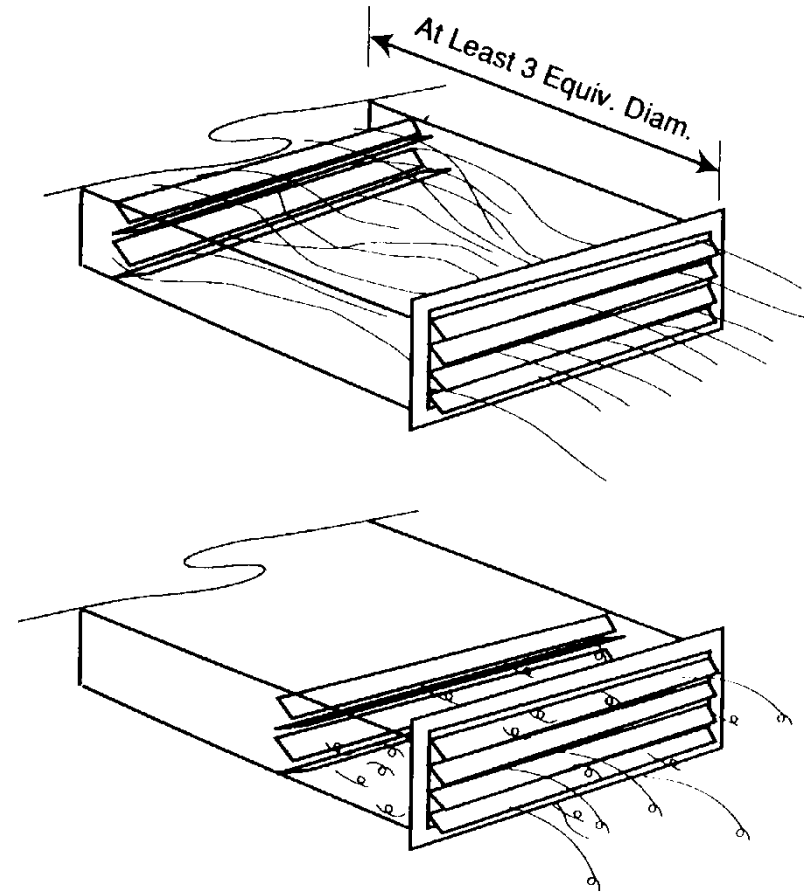
Storage

Terminal Units

- Over an acoustic ceiling, select radiated sound to be 5 NC points lower than the room RC
- Over a drywall ceiling, radiated sound can be 5 NC points higher than the room RC
- Duct taps downstream of box should be at least 4 feet from the unit outlet

Dampers

- Balancing dampers should be **3 or more duct diameters** away from collar or grille
 - Up to 5 dB increase if within 3 diameters if wide open
 - 15-40 dB increase if damper is half open



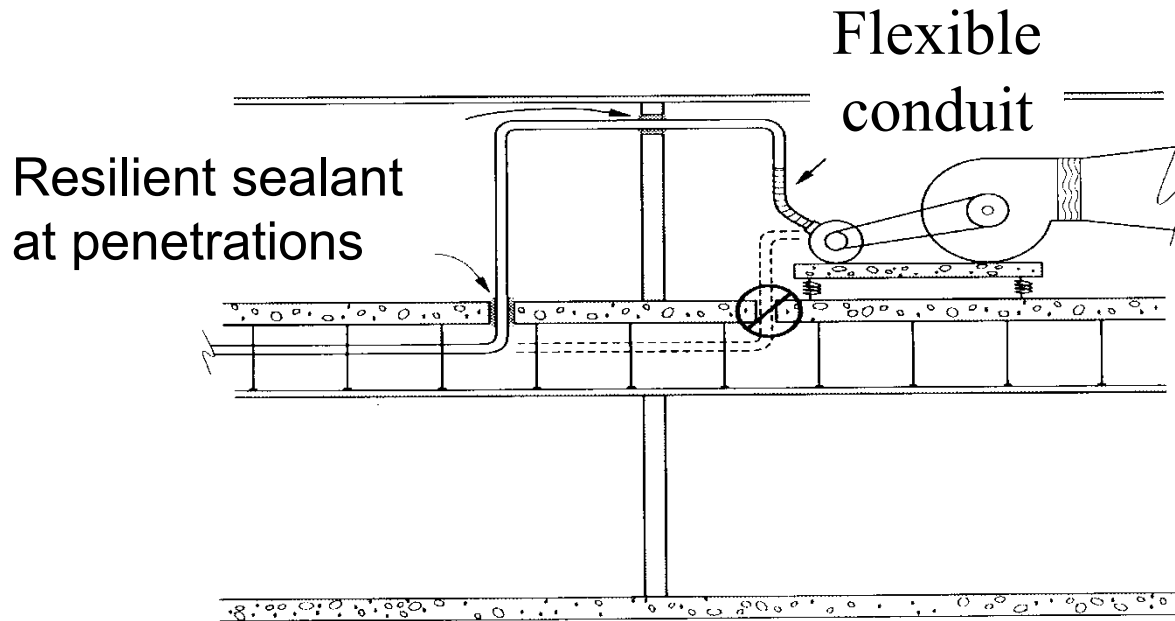
source: ASHRAE *A Practical Guide to Noise And Vibration Control*

Dampers

- Effect on noise of throttling a balancing damper that is integral to the diffuser
- Locating a damper near the branch start is preferable to locating in the diffuser neck

Effective Opening	Increase in Noise Level
100%	0 dB
82%	5 dB
70%	8 dB
50%	16 dB

Wall & Floor Penetrations



- Locate penetrations in less noise sensitive spaces if possible
- Proper sealing of penetrations through partitions

Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4